

Non-Functional Test Cases: Usability Testing

Library Management System

Test Design Team (Non-Coders/Specification Experts)

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Note: These test cases are designed from a black-box, user-centric perspective based on the SRS and the testing plan. They are intended to be executed with representative users (Librarians and Members) , potentially using a think-aloud protocol to gather qualitative feedback. The primary goal is to evaluate the system's user-friendliness, efficiency, and overall user experience as outlined in the SRS usability requirements. Actual results and status are to be filled during test execution.

IV.B Non-Functional Test Cases: Usability Testing

Table 1: Usability Test Cases

Test Case ID	Test Title / Heuristic	Test Scenario & Steps	Expected Results / Success Criteria
Sub-Section: Learnability & Intuitiveness			
TC.USABILITY.LEARN.001	First-time Librarian task completion (Add Book)	Provide a test Librarian account to a new user. - Ask the user to log in and add a new book with specific details (Title, Author, Genre), as per UC-002. - Do not provide instructions beyond the initial goal. - Observe the user's path and listen to their thought process.	- The user can successfully navigate from the dashboard to the book management section and initiate the "Add Book" function without significant hesitation. - The form fields are self-explanatory and easy to fill out. - The user completes the task in a reasonable amount of time with a high degree of confidence.
TC.USABILITY.LEARN.002	First-time Member task completion (Borrow Book)	Provide a test Member account to a new user. - Ask the user to log in, find a specific book, and borrow it, as per UC-003. - Observe the user's search and borrowal process.	- The user can easily locate the book list and understand how to initiate a borrowal request. - The system's workflow for borrowing a book feels logical and straightforward. - The user expresses understanding of the outcome (i.e., that they have successfully borrowed the book).
Sub-Section: Efficiency of Use			
TC.USABILITY.EFF.001	Efficiency of bulk user creation for Librarian	Ask an experienced Librarian user to add five new Member accounts sequentially, as per UC-004. - Time the user. - Ask the user about the repetitiveness and ease of the task.	- The process does not require an excessive number of clicks or redundant data entry for each new user. - The user feels the process is efficient for handling multiple entries. - The system loads quickly between user additions, not impeding the workflow.
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Test Case ID	Test Title / Heuristic	Test Scenario & Steps	Expected Results / Success Criteria
TC_USABILITY_EFF_002	Member's efficiency in viewing borrowal history	Ask a Member to log in and check the status of a previously borrowed book, as per UC-005. Ask them to find their full borrowal history.	- The "Borrowal History" or equivalent section is quickly accessible from the main navigation/dashboard. - The history list is clearly presented and easy to scan for key information (book title, due date, status). - The user does not have to navigate through multiple irrelevant pages to find the information.
Sub-Section: Navigation & Consistency			
TC_USABILITY_NAV_001	Consistent UI layout and controls	Ask a user (either role) to navigate through all major sections of the application: - Dashboard (Librarian) - Book Management - Author Management - Genre Management - User Management (Librarian) - Borrowal Management	- Navigation menus, button styles, search bar placement, and table layouts are consistent across all pages. - Terminology used for actions (e.g., "New", "Edit", "Delete") is uniform throughout the application. - The logo or home link consistently returns the user to their main landing page (Dashboard for Librarian, Books for Member).
TC_USABILITY_NAV_002	Intuitive information architecture	Present a user with a task that spans multiple entities, e.g., "Find all books by a specific author and see if any are currently available." Observe how the user attempts to complete this task.	- The user can logically deduce how to move between the 'Authors' list and the 'Books' list, or can use search/filter functions effectively. - The system provides clear pathways or links between related entities (e.g., clicking an author's name on a book page takes you to the author's detail page).
Sub-Section: Feedback & Error Handling			
TC_USABILITY_FEEDBACK_001	Clarity of success and failure messages	Have a Librarian perform a series of actions: - Successfully delete a genre. - Attempt to delete an author who is currently linked to a book (should fail due to referential integrity). - Successfully update a user's phone number.	- The system provides clear, immediate, and easily noticeable feedback for both successful and failed actions (e.g., "Genre deleted successfully," "Cannot delete author: Author is linked to existing books"). - Confirmation dialogs for destructive actions like deletion are clear and prevent accidental data loss. - Error messages are constructive and tell the user <i>*why*</i> an action failed, not just <i>*that*</i> it failed.
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Test Case ID	Test Title / Heuristic	Test Scenario & Steps	Expected Results / Success Criteria
TC_USABILITY_FEEDBACK_002	Validation feedback on forms	Have a user attempt to submit a form with invalid data. • Example: Submit the "Add User" form with an invalid email format and a blank password field.	• The system prevents form submission. • Clear, inline validation messages appear next to the specific fields that are incorrect (e.g., "Please enter a valid email address," "Password is required"). • The user can easily correct the errors without having to re-enter all the data.
Sub-Section: Responsiveness & Aesthetics			
TC_USABILITY_AESTHETICS_001	Visual clarity and responsive design	Access the application on different screen sizes. • Standard desktop monitor. • Tablet (portrait and landscape). • Mobile phone (portrait).	• The layout adapts gracefully to different screen sizes without horizontal scrolling or overlapping elements. • Font sizes remain readable and interactive elements (buttons, links) are easily tappable on touch screens. • The overall aesthetic is clean, modern, and uncluttered, contributing to a pleasant user experience.